

● HARD

● ALGEBRA

Linear equations in one variable

01

10 questions · ~15 min

Q1

ALGEBRA

$$-49x = -98x$$

How many solutions does the given equation have?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

Q2

ALGEBRA

Each side of a 30-sided polygon has one of three lengths. The number of sides with length 8 centimeters cm is 5 times the number of sides n with length 3 cm. There are 6 sides with length 4 cm. Which equation must be true for the value of n ?

- A. $5n + 6 = 30$
- B. $6n + 6 = 30$
- C. $8n + 3n + 4n = 30$
- D. $85n + 3n + 46 = 30$

Q3

ALGEBRA

$$-3x + 21px = 84$$

In the given equation, p is a constant. The equation has no solution. What is the value of p ?

- A. 0
- B. $\frac{1}{7}$
- C. $\frac{4}{3}$
- D. 4

Q4

ALGEBRA

How many solutions does the equation $1015x - 9 = -156 - 10x$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q5**ALGEBRA**

Hector used a tool called an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours of using the auger, 19,350 bushels of corn remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels of corn remain in the bin?

- A. 3
- B. 7
- C. 8
- D. 12

Q6**ALGEBRA**

A factory makes 9-inch, 7-inch, and 4-inch concrete screws. During a certain day, the number of 9-inch concrete screws that the factory makes is 5 times the number n of 7-inch concrete screws, and the number of 4-inch concrete screws is 22. During this day, the factory makes 100 concrete screws total. Which equation represents this situation?

- A. $95n + 7n + 422 = 100$
- B. $9n + 7n + 4n = 100$
- C. $5n + 22 = 100$
- D. $6n + 22 = 100$

The equation $9x + 5 = a(x + b)$, where a and b are constants, has no solutions. Which of the following must be true?

I. $a = 9$

II. $b = 5$

III. $b \neq \frac{5}{9}$

A. None

B. I only

C. I and II only

D. I and III only

Q8

GRID-IN

ALGEBRA

What value of t is the solution to the equation $0.8t - 0.46 = 8t - 0.001 + 1.9$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
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1	1	1	1
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2	2	2	2
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3	3	3	3
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4	4	4	4
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5	5	5	5
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6	6	6	6
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7	7	7	7
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8	8	8	8
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9	9	9	9
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Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN ALGEBRA

If $5 - 72 - 4x = 16 - 82 - 4x$, what is the value of $2 - 4x$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both.

Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If x is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

- A. $5x$
- B. $10x$
- C. $2x + 20$
- D. $10x + 20$